

What a Cross-Model Fixed-Point Census Can and Cannot Arbitrate About Repetition

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Abstract

Two accounts of neural text degeneration coexist. One locates the cause in the training data: repetition in the corpus produces repetition in the output, established by training models on repetition-sorted data. The other locates it in the trained network, in copying circuits and repetition features. The accounts have never been arbitrated across a broad cohort of *pretrained* models, because the causal work necessarily trains its own. We report an observational cross-model measurement in a different currency: the fixed-point structure of a model’s own short-window argmax map, censused from 96 random two-token starts over 17 off-the-shelf models, with the four-way structural class stable across census seeds on 17 of 17. Three exhibits. At fixed corpus (The Pile) and fixed scale, the class is *not determined*: across two size-matched tiers, *pythia* is a funnel while RWKV, Mamba and a second transformer family are not, and both families hold their class across roughly an order of magnitude of scale. Six of seven models in that ladder reach the *same* endpoint token, and the models that concentrate on it most strongly are among those that never stay there — so what varies is not where trajectories go but whether the destination self-continues. A corpus manipulation available off the shelf, the deduplicated Pythia suite, does not change the class. And the corpus-side *inflow* term proposed for this phenomenon does not select our endpoints once frequency is controlled, on English and on three further languages. We are explicit about what this cannot do: it is observational, it cannot refute a training intervention, and no second funnel family at fixed corpus was within reach, so a recipe idiosyncratic to one model suite is not excluded.

1 Introduction

1.1 Two accounts, and why they have not been arbitrated

Neural language models fall into repetitive loops under maximum-a-posteriori decoding, and there are two families of explanation for it.

The data side. Repetition in the training text produces repetition in the output. [Fu et al. \[2021\]](#) derive this from corpus bigram structure, attributing degeneration to *high-inflow* words — those whose probability sum over all preceding words is large — and state the thesis plainly: the repetition problem is caused by the language itself. The empirical champion is [Li et al. \[2023\]](#), who train GPT-2 on repetition-sorted shards of five datasets, report a strong correlation between repetition rates in training and generated text, and demonstrate causation by dropping out attention to repetitive words during training. They further argue that the high-inflow account, the likelihood-objective account and the self-reinforcement account all reduce to one factor: penalizing repetitions in data.

The weights side. Repetition is a property of the trained network. Copying circuits are the named mechanism, and the two sides have been connected: [Hernandez et al. \[2022\]](#) show that repeated training data disproportionately damages induction heads, and [Aoyama and Wilcox \[2026\]](#) show that surface bigram repetition frequency governs whether induction heads form at all.

The gap is the cohort, not the connection. We do not claim the accounts cannot see each other — they cite each other and cross repeatedly. What is absent is a broad cohort of *independently pretrained* models measured on a common readout. The causal work necessarily trains its own models, which is what makes it causal and also what makes it few-model and single-recipe. [Li et al. \[2023\]](#) say the remaining gap themselves: the model architecture and size may also contribute, but the two factors have not been quantitatively evaluated.

1.2 What this paper is, stated before the results

This is **observational cross-model evidence in a currency different from a training intervention**. It cannot refute [Li et al. \[2023\]](#): they manipulated training data and measured the consequence, and we did not. Our claim is narrower and of a different kind — that on a task-free structural readout, holding corpus and scale fixed does not fix the outcome.

We are equally explicit about the sharpest limit, and it goes here rather than in a limitations paragraph. In our cohort exactly one model family funnels at fixed corpus. **We could not obtain a second funnel family at fixed corpus**, so the possibility that the effect is idiosyncratic to one training recipe is *not excluded* by anything below.

2 Setup

The readout, and what is already published. Iterate the model’s own two-token conditional deterministically, $x_{t+1} = \arg \max_x p(x \mid x_{t-1}, x_t)$, from 96 random two-token starts, and census where trajectories land. The readout is the *fixed-point fraction* ϕ , the proportion of starts terminating in a token that maps to itself. **This map, and the contrast between a model with a dominant attracting fixed point and one without, are established in our own earlier work** [[Vera Zúñiga, 2026](#)], which identifies the attracting fixed point of the argmax map and contrasts a model sending most starts to the newline token against one with no such point. We claim neither here. What is new is the scale of the census, the class taxonomy and its stability, and the corpus-versus-weights attribution the census makes possible.

Classes. Trajectories are classified FUNNEL (one dominant attracting fixed point), NONE (cycles or wandering), FRAGMENTED (many small basins) and BORDERLINE, under thresholds fixed before the data. The funnel geometry — many states feeding one self-continuing token — was derived theoretically by [Fu et al. \[2021\]](#), whose inflow analysis explains why trajectories concentrate; the classes here are that geometry measured on a model’s own conditional rather than on corpus counts.

Stability, which licenses everything else. Across 17 off-the-shelf pretrained models censused at two independent seeds, the class is stable on 17 of 17. The modal endpoint token is stable on 15 of 17; the two exceptions are reported and excluded from every comparison rather than assigned a class.

tier	model	family	ϕ	modal share	modal endpoint
400M	pythia-410m	GPTNeoX	0.458	0.453	\n
400M	rwkv-4-430m-pile	RWKV	0.010	0.521	time
400M	mamba-370m-hf	Mamba	0.010	0.823	first
150M	pythia-160m	GPTNeoX	0.432	0.432	\n
150M	gpt-neo-125m	GPTNeo	0.052	0.464	side
150M	rwkv-4-169m-pile	RWKV	0.000	0.474	\n
150M	mamba-130m-hf	Mamba	0.000	0.573	The

Table 1: Two size-matched tiers, every model trained on The Pile. Within a tier the largest and smallest parameter counts are within $1.16\times$ (400M) and $1.35\times$ (150M) of each other, a bound fixed before the runs. Both census seeds agree on the class for all seven. Source: F178, results/size_matched_pile.json.

Why this readout and not a generation statistic. Nothing is sampled and nothing is generated. There is no decoding temperature, no prompt, and no text to score. That makes the measurement cheap enough to run over many models, and it makes it a property of the conditional rather than of a generation procedure — but it also means our quantity is not the rep- n of the literature, and no result here transfers to that quantity without an argument we do not make.

3 Related work: an adjacent measurement, and what separates it

The dynamical-systems framing of degeneration is not ours and we do not claim it. [Du and Tanaka-Ishii \[2025\]](#) describe degeneration as a collapse from a higher-dimensional trajectory onto a lower-dimensional attractor, borrowing explicitly from chaos theory, and measure a degeneration-detecting dynamical property — the correlation dimension of next-token log-probability trajectories — across several independently trained families. That is the closest published object to the present one, and the differences are worth stating precisely rather than leaving implicit. Their state space is continuous log-probability vectors seeded from real text; ours is a deterministic map over discrete token space seeded from random pairs. Their quantity is a dimension estimated over a trajectory; ours is the fraction of starts reaching a token that maps to itself. Neither measurement is a special case of the other, and the vocabulary they use predates and is independent of our use of it.

Same-corpus, size-matched comparisons of pretrained models across architecture families are likewise an established design rather than a new one, and we adopt it: [Michaelov et al. \[2024\]](#) compare Pythia, RWKV and Mamba checkpoints trained on The Pile at matched weight classes, and [Wang et al. \[2025\]](#) compare Pythia against Mamba at matched size on induction circuits. Our contribution on that axis is a readout, not a design. Both report substantial cross-architecture *similarity*, and [Michaelov et al. \[2025\]](#) report consistency of behavioural phases across architecture, training data and scale. §4 is therefore a counterexample on a readout those studies do not use, and is presented as one.

4 E3: at fixed corpus and fixed scale, the class is not determined

Table 1 holds the corpus fixed and the weight class fixed, and the class still varies. `pythia` is a funnel in both tiers; `RWKV`, `Mamba` and `gpt-neo` are not.

family	size	class	ϕ	modal share	modal endpoint
Pythia	70M	FUNNEL	0.802	0.792	\n
Pythia	160M	FUNNEL	0.432	0.432	\n
Pythia	410M	FUNNEL	0.458	0.453	\n
Pythia	1000M	FUNNEL	0.365	0.406	\n
GPT-Neo	125M	NONE	0.052	0.464	side
GPT-Neo	1300M	NONE	0.000	0.536	\n
GPT-Neo	2700M	NONE	0.036	0.594	\n

Table 2: Both families hold their class across scale on one corpus: Pythia over a $14\times$ span, GPT-Neo over $22\times$. No model is class-unstable across seeds. Source: F179, `results/family_scale_ladder.json`.

It is not a scale effect, and that was the pre-registered risk. An earlier version of this comparison used `gpt-neo-2.7B` against `pythia-410m`, which confounds family with a $6.6\times$ size gap. The decisive cell was registered in advance: `gpt-neo-125m` against `pythia-160m`, $1.28\times$ apart on the same corpus, with the condition that if they landed in the same class the claim would be withdrawn. They do not — $\phi = 0.052$ against 0.432 , both stable across seeds.

We do not say “architecture”, and the reason is in the table. `gpt-neo` is a transformer, from the same laboratory, trained on the same corpus, and it sits with the recurrent models rather than with `pythia`. The split is therefore not transformer-versus-recurrent, and naming it after architecture would be reading the table wrongly. What the evidence supports is the weaker and more precise statement: **at fixed corpus and fixed scale the class is not determined**, and something on the weights side that we have not identified decides it.

The split is a stable family property, not two unrepresentative checkpoints. Table 2 extends each family across scale. Neither ladder breaks.

The sharpest observation is in the endpoint column. Six of the seven models in Table 2 reach the *same* endpoint token, the newline — every Pythia, and GPT-Neo at 1300M and 2700M. GPT-Neo concentrates on it *more* strongly than some Pythias do: modal share 0.536 and 0.594 against `pythia-160m`’s 0.432 . Yet GPT-Neo’s ϕ is 0.000 and 0.036 while every Pythia exceeds 0.36 . The same holds inside Table 1 at matched size: `rwkv-4-169m-pile` and `pythia-160m` share the newline endpoint at nearly the same concentration, 0.474 against 0.432 , with ϕ of 0.000 against 0.432 .

Where trajectories go is shared; whether the destination self-continues is not. The models that funnel hardest onto the newline are among those that never stay on it. This is the decomposition the rest of the paper is organised around, and it is visible at fixed corpus, at fixed scale, and across an order of magnitude.

5 E1: an off-the-shelf corpus manipulation that does not change the class

The Pythia suite was released in duplicate, one copy trained on The Pile and one on a near-deduplicated copy, explicitly so that deduplication could be studied [Biderman et al., 2023]. That

is the nearest thing to the data-side manipulation available without training anything.

Both members remain FUNNEL: $\phi = 0.458$ for `pythia-410m` and 0.427 for `pythia-410m-deduped`, with the same modal endpoint token.

This pair does not differ only in deduplication, and the caveat is not incidental. The deduplicated Pile is approximately 207B tokens while both suites are trained to approximately 300B [Biderman et al., 2023], so the deduplicated models run about 1.45 epochs and re-encounter a large fraction of their corpus a second time. **In a paper about repetition this matters:** the manipulation re-introduces the very variable the data-side account identifies as causal. We therefore do not present this as a clean corpus manipulation. It is a null on a readout with no prior expectation of invariance, from a pair whose difference is smaller than its label suggests.

And a null here is weaker than it looks for a second reason. Hernandez et al. [2022] show that damage from repeated data is strongly non-monotonic — a double-descent, concentrated in a particular range of repetition frequency. If the Pile’s duplication does not fall in that range, the data-side account predicts no effect either, and our null is consistent with both accounts rather than discriminating between them. We report E1 because it is the manipulation the field would ask for, and we report what it does and does not establish.

6 E2: the corpus-side inflow term does not select the endpoints, and this agrees with the data side

Fu et al. [2021] attribute degeneration to high-inflow words, where the inflow of a token is the probability sum over all its predecessors. If the endpoints our maps reach are the tokens that account predicts, they should be high-inflow ones.

On an uncontrolled reading they are: 12 of 13 readable models have an endpoint at or above the 90th inflow percentile, median 99.87. That reading is an artefact of frequency. Fu et al. [2021] themselves distinguish the two, arguing that it is not the high-frequency words but the high-inflow words that lead to repetition, so we compare each endpoint against the 50 tokens nearest it in log-frequency. Against that control the endpoint beats its matched peers in 1 of 13 models, with a median matched percentile of 32.0 against 50 by construction.

The result is not an artefact of measuring every model on English. Three models whose endpoints are not English were measured on their own languages at matched corpus size: 6.0 on Spanish, 16.0 on Korean and 96.0 on Japanese. Two of the three were unmeasurable in English — their endpoint tokens occur zero times in an English corpus — and both land far below the null on the right corpus.

This agrees with the data-side account rather than opposing it. Li et al. [2023] already subsume the inflow account into repetition-in-training-data by a controlled experiment: merging only the repetitive high-inflow pairs, a small fraction of training words, matches the full high-inflow method, while merging random high-inflow pairs of the same size does not. Inflow was demoted by the data side itself. Likewise Michaelov et al. [2025] find that unigram frequency dominates a large share of word-level behavioural variance across pretrained models. **E2 is therefore a consistency check, not a refutation:** our maps funnel to *common* tokens rather than to high-inflow ones, which is what both of those results predict.

7 Limits

The currency gap, first because it is the largest. Li et al. [2023] manipulated training data and measured the consequence. We measured pretrained models and compared them. An observational contrast across models cannot refute a controlled intervention, and nothing here should be read as doing so. Where our results bear on theirs at all, as in §6, they agree.

One funnel family. In our cohort, at fixed corpus, exactly one family funnels. Four Pythia checkpoints funnelling and six non-Pythia checkpoints not funnelling is equally consistent with a property of that training recipe as with anything more general. Distinguishing those requires a second funnel family at fixed corpus; the obvious public candidate on The Pile exceeded the memory available to us, and we did not obtain one. **This is the single largest weakness of §4 and we do not think it can be argued away.**

What “not determined” does and does not mean. §4 holds corpus and parameter count fixed. It does not hold fixed the training schedule, the optimiser, the data order, the tokenizer, or the many choices that differ between independently trained families. A difference under those conditions means *not explained by corpus or scale*. It does not identify what does explain it, and we make no attempt to.

Readout scope. ϕ is a property of a deterministic two-token map. It is not rep- n on generated text, and the relationship between them is unestablished. A model with no fixed points in this census may still degenerate under sampling, and we have not checked.

Statistical refusals, recorded before the numbers. No p -values are reported. The cohort is 17 models and the exhibits rest on two families; a significance test at that n could not fail informatively. No rank correlations are computed between class and scale. Where a comparison could not be made cleanly — two models whose modal endpoint is unstable across seeds, models whose endpoints do not occur in a given corpus — the models are named and excluded rather than assigned a value.

8 Conclusion

On a deterministic, task-free readout of a model’s own conditional, holding the training corpus fixed and the parameter count fixed does not fix the fixed-point structure. Two families trained on the same corpus keep opposite classes across an order of magnitude of scale, and a corpus manipulation released expressly to study deduplication does not move the class either — though that pair is confounded by a second pass over much of its data, and we say so. Meanwhile the corpus-side term proposed for this phenomenon does not select the endpoints once frequency is controlled, in four languages, which is what the data-side literature now predicts for itself.

The observation we would most like tested by others is the decomposition in §4: across families, scales and corpora, models agree remarkably about *which* token trajectories arrive at and disagree about whether that token maps to itself. Six of seven models in one ladder land on the newline; the ones that concentrate on it most strongly are the ones that never stay. If that survives a cohort with more than one funnel family in it, it is a fact about language models that neither account currently explains.

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